

Session Objectives

1. the students will be able to learn the various types and classifications of scoliosis.
2. the students will be able to learn the standard radiological diagnostic method for assessing severity of scoliosis.
3. the students will be able to learn the treatment options for scoliosis.
4. the students will be able to learn the treatment outcomes for scoliosis.

Scoliosis - Classifications

- **Reversibility**
 - **Functional or structural – Structural** curves do not go away with rotational, side-bending, or forward bending motions. **Functional** curves, however, do go away with these motions.
 - Can have patient bend forward until maximal rib hump is visible, then ask patient to swing upper body right then left. Look for reduction of the rib hump with this action.

Scoliosis - Classifications

- **Infantile Idiopathic Scoliosis** –
 - Possibly due to supine and slight side-lying positioning, allowing gravity to cause plastic deformation of the immature thorax. Ideal treatment is prevention, by lying the kids prone (opposite of back to sleep recommendations)
 - Treatment with well fitting orthoses/ casting esp. in early period of skeletal growth.

Scoliosis - Classifications

- **Juvenile Idiopathic Scoliosis**
 - Between 4 and 10 years of age.
 - More progressive than adolescent idiopathic scoliosis
 - <20 degrees – observation with examination and PA x-rays every 4 to 6 months.
 - With evidence of progression (5-7 degrees), bracing is indicated.
 - Variable success of nonoperative treatment.
 - Crankshaft phenomenon – use anterior and posterior fusion to avoid this.
 - Subcutaneous Harrington Instrumentation to decrease amount of shortening (avg. = 4.5 cm if fusion had been done)

Scoliosis - Classifications

- **Adolescent Idiopathic Scoliosis (AIS)**
 - Hereditary predisposition with multifactorial causes. Association with hypokyphosis, possible CNS defect, increased calmodulin, etc.
 - + genetic contribution supported by high concordance of the incidence and severity of scoliosis seen in monozygotic twins, as compared with dizygotic twins.
 - Patients with AIS also often have a sibling or parent with scoliosis.
 - Genetic loci for AIS mapped to chromosomes 17 and 19, with an unclear inheritance pattern.

Scoliosis - Classifications

- Adolescent Idiopathic Scoliosis, cont.
 - Progression associated with:
 - Girls > boys;
 - premenarchal;
 - Risser sign of 0;
 - Double > single curves;
 - Thoracic > lumbar curves;
 - severe curves.
 - Measure height of patient standing and sitting

Scoliosis - Classifications

- **Location**

- **Double Major Scoliosis** – most frequent is thoracic and lumbar; balanced curves; crossover regions subject to degeneration.
- **Single Thoracic Scoliosis** – second most common; if progressive, can affect heart and lungs; quite noticeable.
- **Single Lumbar Scoliosis** – third most common; associated with arthritic change.
- **Junctional Thoracolumbar Scoliosis** – not a common scoliosis; a longer curve; overstresses the spine; arthritic changes.

Scoliosis - Presentation

- Patient may present to pediatrician, because the school nurse noted an abnormality during a scoliosis school screening.
 - Pediatrician should be ready to evaluate the patient for scoliosis, whether noted incidentally, discovered through a school screening program, or when the patient or parents express concern about scoliosis.
 - A spinal examination is important especially around age 10, just before the growth spurt of puberty.

Scoliosis - Evaluation

- Do a careful physical examination with spinal examination
 - Observe how arms are hanging by the body
 - Check levelness of occiput, shoulders, iliac crests, PSIS, PIIS, and greater trochanters.
 - Measurement of leg length
 - **Patient forward bending** (a.k.a. **the Adams forward bend test**) to observe for **Rib Hump**. Does it go away with sidebending to the side of the rib hump?
 - Run fingers along spinous processes.
 - Any lower extremity dysfunctions?

Scoliosis - Evaluation

- Do a careful physical examination with spinal examination
 - Use of a **scoliometer**:
 - Essentially is a carpenter's level
 - **Measures the angle of trunk rotation**
 - How to use:
 - The patient is forward bent, for the Adams forward bend test. Then, run the scoliometer along the patient's spine from cephalad to caudad. If a rib hump is present, the scoliometer ball will **deviate** away from the midline of the apparatus.
 - **Scoliometer degrees are NOT equivalent to the Cobb angle degrees.**
 - **General guideline is:** angle of trunk rotation of 7 degrees correlates with a Cobb angle of 20 degrees

Scoliosis – Radiographic Evaluation

- Radiology – **standardized standing postural x-ray images**
 - Before sending patient, treat patient with OMT to get good spinal mobilization.
 - Looking for any bony pathology, sacral base unleveling, femoral head/ilic crest levelness, and any scoliotic curves – type and severity.
 - **Cobb Method**

Scoliosis – Radiographic Evaluation

- Radiology – **standardized standing postural x-ray images**
 - Is done to **confirm** presence of scoliosis, to **evaluate** the **type** (congenital, neuromuscular, idiopathic) and **severity**, and to **assess the skeletal maturity**, in relation to risk for progression.

Scoliosis – Radiographic Evaluation

- Radiology – **standardized standing postural x-ray images**
 - **Indications:**
 - **Scoliometer reading $>$ or $=$ to 7 degrees** (may need treatment)
 - Presence of an **obvious spinal curve**
 - Thoracic or lumbar **asymmetry** seen on physical examination **in skeletally immature children, + family history of scoliosis**
 - To **monitor progression** in patients already diagnosed with AIS

Scoliosis – Radiographic Evaluation

- Radiology – **standardized standing postural x-ray images**
 - **Initial radiographic evaluation for scoliosis:**
 - Standing, full length
 - Posteroanterior (PA) and lateral views
 - The PA view minimizes radiation to breasts and thyroid
 - As if looking at patient from behind, patient's heart is to the examiner's left
 - From C7 to the sacrum and iliac crest
 - All exposed on a single long film cassette (send patient to a facility where this is done, versus obtaining suboptimal x-rays)

Scoliosis – Radiographic Evaluation

- Radiology – **standardized *standing* postural x-ray images**
- **Initial radiographic evaluation for scoliosis, cont.:**
 - DO NOT have radiographs taken in the *supine* position, since this decreases the effect of gravity on the spinal curves, leading to underestimation of the spinal curve(s)
 - ***Look for air-fluid level in the stomach, which confirms that x-ray was done in standing position***
 - For a true leg length discrepancy, place the appropriate size block/lift under the shorter leg before the radiographic procedure.
 - ***To determine the pure scoliotic curve, without the additional compensation added in this case***
 - Lateral bending films
 - ***These are to be obtained by the operating surgeon as necessary, and can show the potential for surgical correction***

Scoliosis – Radiographic Evaluation

- For **bone age/skeletal maturity** in adolescents with a ***rapidly progressing curve***:
 - Check x-ray of iliac apophysis (***Risser sign***)
 - Check x-ray of hand (***Sanders Skeletal Maturity Staging System***)
 - ***Also, can check the following (FYI):***
 - Check x-ray of ***epiphyses of distal radius, ulna, and small bones of the hand*** (the method of ***Tanner and Whitehouse***)
 - Check x-ray of ***elbow*** (assessment of ossification centers in the elbow)
 - ***Sauvegrain*** method
 - The (simpler) ***olecranon*** method

Scoliosis – Radiographic Evaluation

- For Bone age/skeletal maturity: x-ray of iliac apophysis (**Risser sign**):
 - *The Risser sign is a visual grading of the degree of ossification and/or fusion present in the iliac apophysis:*
 - 0 – no ossification
 - 1 – up to 25% ossification (Risser 1 usually occurs AFTER peak height velocity)
 - 2 – 26% to 50% ossification
 - 3 – 51% to 75% ossification
 - 4 – greater than 76% ossification
 - 5 – Full bony fusion of the apophysis
 - **A lower Risser grade, means that the patient has quite a bit more growth remaining, and is therefore at a greater risk for a scoliotic curve progression**

Scoliosis – Radiographic Evaluation

- MRI indications, cont.:
 - Associated neurologic s/s:
 - Headache, neck pain, absence of abdominal reflex, asymmetric lower extremity atrophy, pes cavus, midline skin lesions (e.g. vascular, pigmentary, hair patch)
 - Abnormal findings on the plain radiographs:
 - Findings correlating with congenital scoliosis
 - Findings suggestive of intraspinal pathology
 - Increased kyphosis

Scoliosis – Radiographic Evaluation

- **Indications for REFERRAL to Orthopedic specialist:**
 - At time of referral, be clear about **which method of measurement was used – Cobb angle or Scoliometer** – since the degree values represent quite disparate values, with variable risks for progression.
 - ***Orthopedic referral is not necessary for incidental findings of scoliosis on a chest x-ray, when the Cobb angle measure is < 10 degrees.***

Scoliosis – Treatment

- **Mild Scoliosis** – OMT, (Konstancin, per Campbell's Orthopedics) exercises, functional orthotics, education
- **Moderate Scoliosis** – OMT, exercises, education, bracing.
 - NOTE: For Mild and Moderate Scoliosis, **particular Isometric Yoga Poses OR the Schroth Method for Scoliosis** (*see CPG for Supplemental information, not required*) could be additional options considered, sometimes in conjunction with bracing, and the other options, as well.
- **Severe Scoliosis** – Surgery and adjunctive treatment as above.

Scoliosis – Treatment

- OMT goal in the treatment of structural scoliosis is to **optimize function in the existing structure, not to straighten the curves.**
- Indirect treatment, fascial release, soft tissue treatment, gentle Balanced Ligamentous/Membranous Tension (BLT, BMT) treatments, to increase general body ROM.
- Abdominal and Psoas muscle strengthening.
- Can treat joint somatic dysfunctions as well.

Scoliosis – Treatment

- For **Adolescent Idiopathic Scoliosis (AIS)**, **the goal** is spinal curve at **skeletal maturity** with a **Cobb angle of 40 degrees or less.**
- If the **Cobb angle at skeletal maturity is 50 degrees or greater**, then these can progress at a rate of **1 degree per year after growth ceases.**
 - E.g. after 30 years → 30 degrees + 50 degrees = 80 degree+ curve.
Association with restrictive lung disease, compromised pulmonary function.

Scoliosis – Treatment

- **Bracing:**

- Indicated for **skeletally immature** patients (**Risser 0 to 2, Sanders Stage 1 to 3**).
 - Not indicated for skeletally mature patients (Risser 3 or higher, Sanders Stage 4 to 8), as they have little growth remaining.
- **Cobb angle is 25 to 49 degrees at the time of presentation**
- **Cobb angle progresses $>$ or $=$ to 5 degrees in any 6-8 month observation period**
- **May prevent curve progression**, potentially reducing the need for surgery
- **Does not correct the curve present at the time of diagnosis**

Scoliosis – Treatment

- **Bracing:**
 - **Contraindicated in patients who have reached skeletal maturity (*Risser 4 to 5, ring apophyses fused*)**
 - **Contraindicated in patients with a Cobb angle $>$ or $=$ to 50 degrees**
 - **Relatively contraindicated in patient's with a thoracic lordosis**

Scoliosis – Treatment

- **Braces**

- **Underarm orthoses** utilized for most curves– **TLSOs (thoraco-lumbar-sacral orthosis)** like the **Boston Brace** are used now.
 - Also: the **Wilmington plastic jacket**, the **Charleston brace**, and the **Providence brace**
 - Prevents progression of larger curves when worn 18 hours/day.
- **Milwaukee Brace** (23 hours/day) **rarely used now**, a.k.a. **CTL SO (cervico-thoraco-lumbar-sacral orthosis, has an under-chin extension)**
 - 50% improvement after 6 months in the brace; but at 5 year follow-up, curve was the same as at diagnosis.

Scoliosis – Treatment

- **Efficacy of bracing – *mixed results*** reported in various reviews of the literature.
- National Institutes of Health “**BrAIST**” trial.
 - The Bracing in Adolescent Idiopathic Scoliosis Trial (BrAIST)
 - Five-year, multicenter (20), prospective, randomized trial: compared bracing and observation in skeletally immature patients
 - Cobb angle between 25 to 40 degrees

Scoliosis – Treatment

- **Bracing:**

- Recommended brace wear of 23 hours per day in the past. Now, 18 hours per day is recommended, however-
 - *BrAIST trial found that wearing the brace approx. 13 hours per day was effective.*
 - *With the exception of patients at Risser Stage 0, who may benefit from more hours wearing the brace.*
 - **Actual usage**, compared to the prior 23 hours that was recommended, **was approximately 65% of prescribed time**
 - Adherence to wearing the brace can be **monitored objectively** with **temperature or pressure sensors** inside the brace
 - Patient **may wear the brace during sports; may remove** for swimming, gymnastics-type sports as needed

Scoliosis – Treatment

- **Bracing:**

- **Check for proper fit of the brace, perform an in-brace radiograph**
 - **Look for the amount of curve correction, goal is 20% reduction of the in-brace Cobb angle → leads to greater chance of success**
 - **Brace treatment is continued until the end of the growth period;**
 - **Girls:** Two years post-menarche, Risser grade 4 or 5
 - **Boys:** Risser grade 5
 - Can either **stop completely, or wean down** to just nighttime usage (orthopedic practitioner clinical choice)
 - **Follow up at six-months post-bracing, and then yearly for a few years.** An initial curve collapse (several degrees) will soon stabilize.

Scoliosis – Treatment

- **Bracing:**
 - **Monitor the brace treatment every 6-8 months both clinically and with follow up radiographs.**
 - **Out of brace radiographs are performed to check for any curve progression**
 - **Progression of the Cobb angle by $>$ or $=$ to 5 degrees during bracing treatment is a poor prognosis for the efficacy of this treatment for the patient, and surgery may be needed in these cases.**

Scoliosis – Treatment

- **Surgery for AIS:**
 - **Primary goal** = utilizing **spinal fusion** for **prevention of curve progression**
 - **Secondary goal** = **partial curve correction**, this is frequently achieved
 - **Indications:**
 - **Skeletally immature patients**
 - **At diagnosis**, patient has **curve with Cobb angle of > or = to 50 degrees.**
 - **Sometimes** skeletally immature patients **with Cobb angle between 40 to 50 degrees *may* need surgery.**

Scoliosis – Treatment

- **Surgery for AIS:**
 - Surgical procedures **utilized:**
 - **Posterior spinal fusion (most common)-**
 - Bone grafting and instrumentation
 - **Instrumentation has evolved** from the Harrington rods (cephalad and caudad spinal connections with hooks), to the modern segmental instrumentation (hooks, screws, wires, rods).
 - Modern instrumentation **provides the orthopedic surgeon with more control for positioning of the spine.**
 - Most important part of the surgery is obtaining **bony fusion**
 - **Autograft or allograft**

Scoliosis – Treatment

- **Surgery for AIS:**

- Surgical procedures, **outcomes:**
 - **95% of cases have successful fusion**
 - **Complications:** implant failure, pseudoarthrosis (failure of fusion), neurologic injury, infection, blood loss
 - **Intraoperative monitoring advances have further decreased the risk of complications**
 - ***Limited evidence*** regarding the ***primary goal of prevention of curve progression*** with the spinal fusion procedure
 - ***The secondary goal of coronal curve correction had variable results*** between 48-67% correction with a posterior fusion, and 71-93% with anterior fusion surgical procedures.

Scoliosis – Treatment

- **Curve progression in AIS:**
 - ***Untreated curves may continue to progress after skeletal maturity***
 - Occurs in 2/3 of patients
 - ***Curves <30 degrees are much less likely to progress***
 - One **50-year follow up** showed patients had **little impairment** other than ***back pain and cosmetic concerns***

Scoliosis – Treatment

- **Curve progression in AIS, cont.:**
 - **Curves treated with bracing and surgery can also progress:**
 - **Bracing treatment**, patients followed for 22 years, average coronal curve progression of **7.9 degrees**
 - **Surgical treatment**, patients followed for 23 years, average coronal curve progression of **3.5 degrees**

Scoliosis – Treatment

- **Scoliosis and back pain:**

- Whether the scoliosis is ***treated or untreated***, there is a **slightly increased occurrence of back pain in adults with AIS**, compared to controls.
 - There ***may be a correlation*** between the ***initial magnitude of the curve*** and ***later back pain complaints***
 - ***Patients with AIS treated with bracing or surgery may also have increased chance of degenerative disc changes.***

Scoliosis – Treatment

- **Scoliosis and pregnancy:**

- A patient's pregnancy ***will not affect*** her scoliotic curve's progression
 - **The risk of progression not affected by:**
 - The number of pregnancies
 - Patient's age at the time of the first pregnancy
 - The curve stability
- The scoliotic curve itself ***does not seem to affect the outcome*** of the pregnancy
 - **No apparent increase in cesarean section deliveries or other health concerns in pregnancy**, as compared to controls.

Scoliosis – Treatment: Summary

- **Generally, in growing patients with a flexible 20-30 degree curve, with documented progression of 5 degrees or more, orthotic treatment is indicated. Adjunctive treatments may be included.**
- **Larger curves of 30-40 degrees in growing children, found upon the initial visit, are treated with orthotics right away.**
- **40-50 degrees – surgery *or may consider orthotics for double major curves***
- **>50 degrees - surgery**

Summary Slide

1. Understanding the various types and classifications of scoliosis and knowing the standard radiological diagnostic method for assessing severity of scoliosis.
2. Understanding the treatment options and outcomes for scoliosis.